

The forward models of the `ode-models` crate

September 17, 2026

What a model provides

Every model of the crate implements one contract, the `Model` trait of `crates/ode-models/src/model.rs`: parameters and pure maps of a state, no state of its own. It provides a time step δt ; the exact discrete flow φ over one step and its inverse φ^{-1} (for the symmetric Gauss–Legendre schemes, one step with $-\delta t$; for the linear chain, a stored matrix); the observation $h(x)$, the discrepancy $d(y, h(x))^2$ the filter multiplies the density by given the observation y of the step, and the innovation $y \ominus h(x)$ (the shortest angular difference for a bearing); the tangents $\partial\varphi$ and ∂h for the Gaussian closure; the periodic components with their intervals (the angles of the pendulums); and whether the flow is autonomous, which lets the filter cache φ^{-1} of its grid. The state is one flat vector, $x \in \mathbb{R}^n$, in the order given with each model. The reference trajectory and its observations $y_n = h(x_n) + \eta_n$ are generated outside the model (`ode-models-spec`), by the same flow. The viewer’s “About this model” button opens this document at the model’s section.

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1 Linear spring chain

N equal masses ρ on a line, body 1 attached to a wall, each body to the next by a spring of stiffness a per unit length, the free end unattached. With $Y \in \mathbb{R}^N$ the positions and $V = \dot{Y}$,

$$M \ddot{Y} + K Y = 0, \quad M = \rho I, \quad K = \frac{a}{\ell} \text{tridiag}(-1, 2, -1) \text{ with last row } (-1, 1),$$

a conservative linear system, $H = \frac{1}{2} V^\top M V + \frac{1}{2} Y^\top K Y$. **State** $x = (Y; V) \in \mathbb{R}^{2N}$. **Time stepping** implicit midpoint: $x_{n+1} = T x_n$ with a constant matrix T , symplectic, exactly invertible (T^{-1} stored), so transport is the linear map $\varphi^{-1} = T^{-1}$.

Observation one position $h(x) = y_i$ (the selected mass), optionally with noise. **Role in the library:** the linear-Gaussian reference — the tracker equals the Kalman filter here, and the grid filter's argmax follows it to node accuracy.

References: [Harmonic oscillator \(Wikipedia\)](#), [Implicit midpoint method \(Wikipedia\)](#).

2 Spring chain with an unknown mass

N bodies on a line, body 1 attached to a wall, consecutive bodies linked by springs of stiffness k ; $y_1, \dots, y_N$ are the displacements from rest. Bodies $1, \dots, N-1$ have the known mass m , the free-end body the unknown mass

$$m_N(\theta) = m_0 2^\theta, \quad \dot{\theta} = 0,$$

θ the log-parameter to estimate (in doublings of m_0 ; positive m_N for every θ , a Gaussian prior in θ is log-normal in m_N). For $N = 1$ the single body is the unknown one. **Dynamics**

$$\dot{y} = v, \quad m_i \dot{v}_i = -(Ky)_i, \quad \dot{\theta} = 0, \quad K = k \operatorname{tridiag}(-1, 2, -1), \quad K_{NN} = k.$$

State $x = (y_1..y_N, v_1..v_N, \theta) \in \mathbb{R}^{2N+1}$. **Time stepping** Gauss–Legendre 4 (symplectic, symmetric: φ^{-1} = one step with $-\delta t$); θ is exactly conserved, energy at fixed θ too.

Conditionally linear: given θ the dynamics are linear in (y, v) — the class where the hybrid grid–Gaussian closure is exact. **Observation** one position y_i . In the filter $q_\theta = 0$ (no diffusion in θ): each θ -slice runs its own chain, the observation alone discriminates them.

References: [Harmonic oscillator \(Wikipedia\)](#), [Rao–Blackwellisation \(Wikipedia\)](#).

3 Double pendulum, point masses

Mass m_1 hangs from a fixed pivot by a massless rod of length ℓ_1 , mass m_2 from m_1 by a rod of length ℓ_2 ; gravity g downward. **State** $x = (q_1, q_2, p_1, p_2)$, q_i the angle of rod i with the downward vertical, p_i its conjugate momentum, $\Delta = q_1 - q_2$:

$$H(x) = \frac{m_2 \ell_2^2 p_1^2 + (m_1 + m_2) \ell_1^2 p_2^2 - 2m_2 \ell_1 \ell_2 p_1 p_2 \cos \Delta}{2m_2 \ell_1^2 \ell_2^2 (m_1 + m_2 \sin^2 \Delta)} \\ + (m_1 + m_2)g\ell_1(1 - \cos q_1) + m_2 g \ell_2 (1 - \cos q_2) + m_2 g \ell_2.$$

Hamilton's equations $\dot{q} = \partial_p H$, $\dot{p} = -\partial_q H$; chaotic at large amplitude. **Time stepping** Gauss–Legendre 4 (2-stage implicit Runge–Kutta, symplectic; Newton with the analytic Jacobian), symmetric so φ^{-1} = one step with $-\delta t$.

Observation the tip position $(x, y) = (\ell_1 \sin q_1 + \ell_2 \sin q_2, -\ell_1 \cos q_1 - \ell_2 \cos q_2)$, a vector observation of the two angles. The angles are periodic on $(-\pi, \pi]$: the filter uses periodic boundary conditions in q_1, q_2 .

References: [Double pendulum \(Wikipedia\)](#), [Gauss–Legendre methods \(Wikipedia\)](#).

4 Double pendulum, rigid rods

Two identical uniform rods of mass m and length ℓ , pinned end to end below a fixed pivot; gravity g downward (the *swaptube* model). Moment of inertia $I = m\ell^2/3$ about a rod's centre. **State** $x = (q_1, q_2, p_1, p_2)$, angles with the downward vertical and conjugate momenta, $\Delta = q_1 - q_2$:

$$H(x) = \frac{6(p_1^2 + 4p_2^2 - 3p_1p_2 \cos \Delta)}{m\ell^2(16 - 9\cos^2 \Delta)} + \frac{1}{2}mg\ell(3(1 - \cos q_1) + (1 - \cos q_2)),$$

$$\begin{aligned} \dot{q}_1 &= \frac{6}{m\ell^2} \frac{2p_1 - 3p_2 \cos \Delta}{16 - 9\cos^2 \Delta}, & \dot{q}_2 &= \frac{6}{m\ell^2} \frac{8p_2 - 3p_1 \cos \Delta}{16 - 9\cos^2 \Delta}, \\ \dot{p}_1 &= -\frac{1}{2}m\ell^2(\dot{q}_1\dot{q}_2 \sin \Delta + 3\frac{g}{\ell} \sin q_1), & \dot{p}_2 &= -\frac{1}{2}m\ell^2(-\dot{q}_1\dot{q}_2 \sin \Delta + \frac{g}{\ell} \sin q_2). \end{aligned}$$

Time stepping Gauss–Legendre 4, symplectic and symmetric (φ^{-1} = one step with $-\delta t$). Released from rest at $(q_1, q_2) = (2.453, -2.7727)$ the tip traces the heart-shaped orbit. **Observation** the tip position; angles periodic on $(-\pi, \pi]$.

References: [Double pendulum \(Wikipedia\)](#), [swaptube \(GitHub\)](#).

5 Kepler problem, softened

A body of unit mass in the field of a fixed centre at the origin, in the orbital plane. **State** $x = (q_1, q_2, p_1, p_2)$, Cartesian position and momentum, with the Plummer-softened potential

$$H(x) = \frac{1}{2}|p|^2 - \frac{\mu}{\sqrt{|q|^2 + a^2}}, \quad \dot{q} = p, \quad \dot{p} = -\frac{\mu q}{(|q|^2 + a^2)^{3/2}}.$$

$a = 0$ is Newtonian gravity (closed ellipses); $a > 0$ (default 0.1) bounds the force by μ/a^2 , so the flow and its Newton solve are defined at every point of the filter's box, which contains the origin; at $|q| \approx 1$ the change is $O(a^2)$. Units: $\mu = 1$ gives the circular orbit of radius 1, speed 1, period 2π ; the orbit of semi-major axis 1 and eccentricity e starts at perihelion $q = (1 - e, 0)$, $p = (0, \sqrt{(1 + e)/(1 - e)})$.

Conserved energy H and angular momentum $L = q_1 p_2 - q_2 p_1$ (a quadratic invariant, conserved exactly by Gauss–Legendre 4; H up to the bounded $O(\delta t^4)$ oscillation of a symplectic scheme).

Observations (one selected): the coordinate q_1 ; the range $|q|$ (the polar angle unobserved: ring-shaped densities in the (q_1, q_2) plane); the bearing $\text{atan2}(q_2, q_1)$, whose discrepancy is the shortest angular difference.

References: [Kepler problem \(Wikipedia\)](#), [Plummer model \(Wikipedia\)](#).

6 Kepler problem with an unknown gravitational parameter

The softened Kepler dynamics with the gravitational parameter as an extra state:

$$\mu(\theta) = \mu_0 2^\theta, \quad \dot{\theta} = 0, \quad \dot{q} = p, \quad \dot{p} = -\frac{\mu(\theta) q}{(|q|^2 + a^2)^{3/2}}.$$

State $x = (q_1, q_2, p_1, p_2, \theta) \in \mathbb{R}^5$; θ counts doublings of μ_0 ($\theta = \pm 1 \Leftrightarrow \mu = 2\mu_0, \mu_0/2$), $\mu > 0$ for every real θ , and a Gaussian prior in θ is log-normal in μ . **Time stepping** Gauss–Legendre 4; θ is exactly conserved (its stage equation decouples), L too, H up to $O(\delta t^4)$.

Identifiability μ sets the period (Kepler’s third law) and is identifiable from position observations; the satellite’s mass would not be, since it cancels from $\ddot{q}$. **Observations** q_1 , range or bearing, all functions of q only. **In the filter** $q_\theta = 0$: each θ -slice runs its own Kepler flow and the observation discriminates the slices; the twin experiment sets $\theta_{\text{true}} = 0.4$, the regime where the q_1 -likelihood is phase-aliased and the Gaussian tracker falls into a secondary basin — the grid’s job.

References: [Kepler problem \(Wikipedia\)](#), [Kepler’s third law \(Wikipedia\)](#), [Plummer model \(Wikipedia\)](#).

7 Lorenz-63

The three-mode truncation of Rayleigh–Bénard convection (Lorenz 1963),

$$\dot{x} = \sigma(y - x), \quad \dot{y} = x(\rho - z) - y, \quad \dot{z} = xy - \beta z,$$

state $(x, y, z) \in \mathbb{R}^3$, parameters σ, ρ, β (defaults 10, 28, 8/3: the strange attractor). Dissipative: $\nabla \cdot f = -(\sigma + 1 + \beta)$, phase-space volume contracts at that constant rate — no Hamiltonian structure, and without model noise the filter density would collapse onto the attractor: the model-noise q_d is what keeps it spread. Fixed points: the origin and, for $\rho > 1$, $C_{\pm} = (\pm\sqrt{\beta(\rho - 1)}, \pm\sqrt{\beta(\rho - 1)}, \rho - 1)$, the centres of the two wings.

Time stepping Gauss–Legendre 4 (A-stable, symmetric: $\varphi^{-1} =$ one step with $-\delta t$). The inverse flow of a dissipative system is expanding: leave a generous margin around the attractor in the filter’s box. **Observation** one coordinate, x , y or z .

References: [Lorenz system \(Wikipedia\)](#), [Lorenz 1963 \(JAS, doi\)](#).

8 Lamppost

A man stands in the plane at (x, y) next to a lamppost at the origin and does not move,

$$\frac{d}{dt}(x, y) = 0, \quad h(x, y) = x^2 + y^2,$$

all that is observed being his squared distance to the lamppost. With the reference at $(0, 1)$ the observation is $y_{\text{obs}} \equiv 1$ and every point of the unit circle explains the data: the value function is flat on the ring $x^2 + y^2 = 1$ and the density is a *ring*, a continuum of maxima. The grid filter shows it; the tracker, one Gaussian, converges to some point of the ring chosen by the prior — the smallest illustration of what the Gaussian closure loses.

The drunkenness is the filter’s model noise q : the diffusion keeps the ring from collapsing to a curve ($q \equiv 0$ sharpens it forever). Optionally the reference itself random-walks (a standard deviation per step), so the ring follows a moving radius. **Flow** the identity, exactly: no time stepping, $\Phi = I$.

References: [Random walk \(Wikipedia\)](#) (the optional walk of the reference; the model itself is the identity).

9 Cardiac cavity (spherical ventricle)

The 0D cardiac cavity of the PhysioBlocks library [1, Appendix C], itself the spherical reduction [2] of the Inria heart model [3], in the variant of [4]. Continuous level only; not implemented in the crate yet.

Kinematics. The wall is a sphere of mid-surface radius R_0 and thickness d_0 at rest; one scalar y is the change of radius, $R = R_0 + y$. With

$$C(y) = \left(1 + \frac{y}{R_0}\right)^2, \quad V(y) = \frac{4}{3}\pi \left(R_0 + y - \frac{d_0}{2} C(y)^{-1}\right)^3, \quad |\Omega_0| = 4\pi R_0^2 d_0,$$

C the fibre component of the right Cauchy–Green tensor, V the cavity volume, $|\Omega_0|$ the tissue volume at rest.

Constitutive functions. Passive elastic energy per unit volume, viscous stress, and the fraction of active tension reachable at a given sarcomere extension e_c :

$$W_e(y) = C_0 \exp\left[C_1(2C(y) + C(y)^{-2} - 3)^2\right] + C_2 \exp\left[C_3(C(y) - 1)^2\right], \quad \Sigma_v(y, \dot{y}) = 2\eta (C(y) + 2C(y)^{-5}) \frac{\dot{y}}{R_0},$$

and $n_0 : \mathbb{R} \rightarrow [0, 1]$, a given piecewise-linear function (Fig. 9 of [1]).

State. $x = (y, v, e_c, K_c, T_c) \in \mathbb{R}^5$: the radius change y and its velocity $v = \dot{y}$, the sarcomere extension e_c (relative sliding of the myosin over the actin filaments, per half-sarcomere length), the active stiffness K_c and the active tension T_c .

Inputs. The cavity pressure $P_V(t)$ (the coupling to the circulation, through the flux $Q = -\dot{V}(y)$ the cavity delivers) and the electrical activation $u(t)$ (positive during depolarisation, negative during repolarisation; a given periodic function, Fig. 8 of [1]). Both are prescribed functions of time here.

First-order system.

$$\begin{aligned} \dot{y} &= v, \\ \rho_0 \dot{v} &= -W'_e(y) - \frac{1}{R_0} \Sigma_v(y, v) - \frac{k_s}{R_0} \left(\frac{y}{R_0} - e_c\right) + \frac{P_V}{|\Omega_0|} V'(y), \\ \mu \dot{e}_c &= k_s \left(\frac{y}{R_0} - e_c\right) - T_c, \\ \dot{K}_c &= -(|u| + \alpha |\dot{e}_c|) K_c + n_0(e_c) K_0 |u|_+, \\ \dot{T}_c &= -(|u| + \alpha |\dot{e}_c|) T_c + K_c \dot{e}_c + n_0(e_c) T_0 |u|_+, \end{aligned}$$

ρ_0 the tissue density, k_s the passive elastic modulus of the sarcomeres (the series elastic element between the tissue strain y/R_0 and the sarcomere sliding e_c), μ the sarcomere viscosity, α the destruction rate per unit extension rate, K_0 and T_0 the maximal active stiffness and tension. The second equation is the momentum balance of the wall (passive elasticity, viscosity, sarcomere force, blood pressure); the third is the force balance in the sarcomere; the last two are the two-moment reduction of the Huxley'57 crossbridge model [3], linear in (K_c, T_c) given $(e_c, \dot{e}_c, u)$: sustained $u > 0$ drives (K_c, T_c) towards $n_0(e_c) (K_0, T_0)$, sustained $u < 0$ towards $(0, 0)$. ([1] uses $\lambda_c = T_c/\sqrt{K_c}$ in place of T_c for its discretisation.)

Observation not defined yet.

References

- [1] D. Chapelle, C. Drieu, F. Kimmig. *PhysioBlocks: A Python Library for Graph-Based Nodal Simulation of Dynamical Physiological Systems*. Preprint, 2026. [hal-05436182v2](https://hal.archives-ouvertes.fr/hal-05436182v2).

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